

Supplementary Information for

Tailoring Cas12a Functionality with a User-Friendly and Versatile crRNA

Variant Toolbox

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Supplementary Table 1 | Strains used in this work.

Strain	Relevant characteristics
<i>E. coli</i> NEB 10-beta	Wild-type strain
<i>E. coli</i> Turbo	Engineered <i>E. coli</i> for plasmid construction
<i>E. coli</i> NEB 10-beta- <i>gfp</i>	Wild-type <i>E. coli</i> NEB 10-beta strain fused with green fluorescent protein encoding gene in genome
<i>E. coli</i> Rosetta	Engineered <i>E. coli</i> for Cas12a expression

Supplementary Table 2 | Primers and probes for RPA and qPCR.

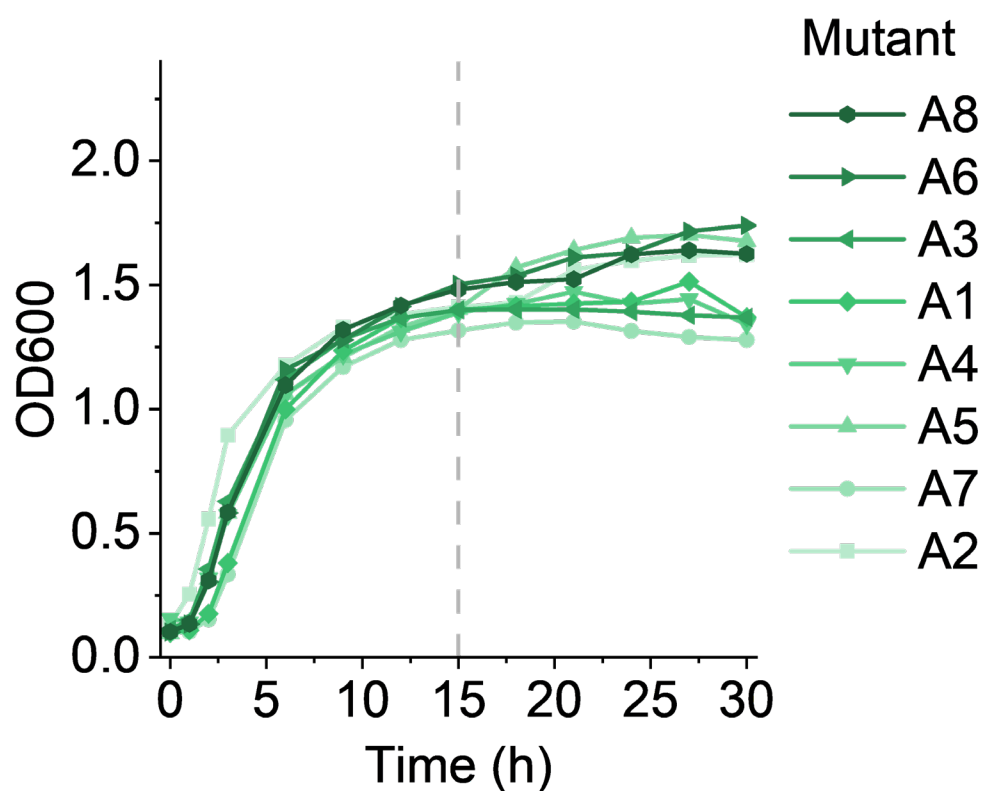
Name	Sequence (5' to 3')	Purpose
P1-RPA-F	CACCAACGAACAGAGCTTAGGTCTCCGCTT	RPA
P1-RPA-R	CTAAACAAGGTTTGGGGACCTTGACTGGAG	
P1-qPCR-F	CCAACCAAACAACAACGTTCA ¹	qPCR
P1-qPCR-R	ACCTTGACTGGAGGCCGTTA ¹	
P1-qPCR-probe	FAM-TCAATCCGAATAACGGTGACTTCTTACCACTG-BHQ1 ¹	

Supplementary Table 3 | Plasmids used in this work.

Plasmid	Description	Resistance	Source or reference
pCRISPRi	Gene repression plasmid, araC-P _{BAD} -dCas12a, P _{J23119} -crRNA	Kan	This study
pBE-APOBEC-	Base editing plasmid, araC-P _{BAD} -APOBEC-dCas12a-UGI, P _{J23119} -crRNA	Kan	This study
pBE-evoCDA-	Base editing plasmid, araC-P _{BAD} -evoCDA-dCas12a-UGI, P _{J23119} -crRNA	Kan	This study
pBE-haA3A	Base editing plasmid, araC-P _{BAD} -haA3A-dCas12a-UGI, P _{J23119} -crRNA	Kan	This study
pCas12a	araC-P _{BAD} -Cas12a	Kan	This study
pKO	Fragment editing plasmid, P _{J23119} -crRNA, homologous arm for 500 bp knockout	Cm	This study
pKIN	Fragment editing plasmid, P _{J23119} -crRNA, homologous arm for 1200 bp knock-in	Cm	This study
pET28-Cas12a	Cas12a expression plasmid	Kan	²
pUC57-P1	Synthetic P1 adhesin gene of <i>Mycoplasma pneumoniae</i>	Amp	This study

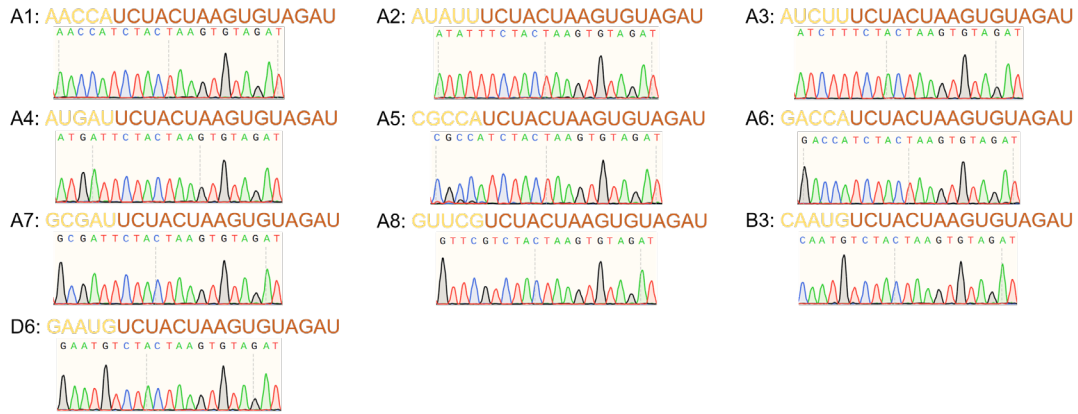
Supplementary Table 4 | Ct values of clinical samples of *Mycoplasma pneumoniae*

Sample	Ct values	Sample	Ct value	Sample	Ct values	Sample	Ct value
1	34.57	11	27	21	29.95	31	30.61
2	33.52	12	25.58	22	27.2	32	30.82
3	34.96	13	27.52	23	33.48	33	24.49
4	32.37	14	33.96	24	34.35	34	30.7
5	32.79	15	30.1	25	29.01	35	31.92
6	23.93	16	28.81	26	25.88	36	28.99
7	31.52	17	27.28	27	28.65	37	34.19
8	33.07	18	28.96	28	26.97	38	23.93
9	27.11	19	34.18	29	31.49	39	33.13
10	27.94	20	32.5	30	33.7	40	29.16

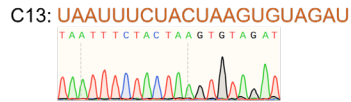


Supplementary Fig. 1 | Growth curve of *E. coli* NEB 10-beta-*gfp* with flanking sequence-mutated crRNAs guiding dCas12a to knock down *gfp*. Source data are provided as a Source Data file.

Flanking sequence-mutated crRNAs (NNNNN_{5'}UCUACUAAGUGUAGAU):



Stem-mutated crRNAs (UAAUU_{5'}NNNNN_{3'}UAAGUNNNNN_{5'}):

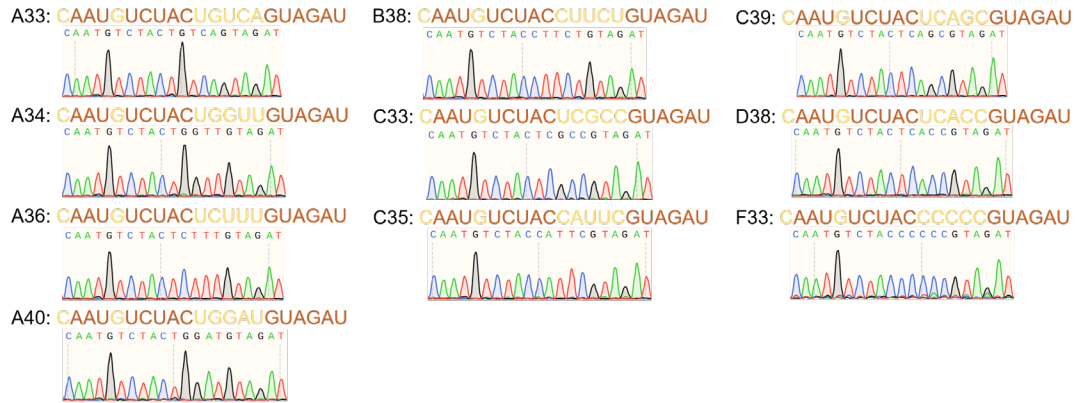


Loop-mutated crRNAs (UAAUUUCUAC_{5'}NNNNN_{3'}GUAGAU):

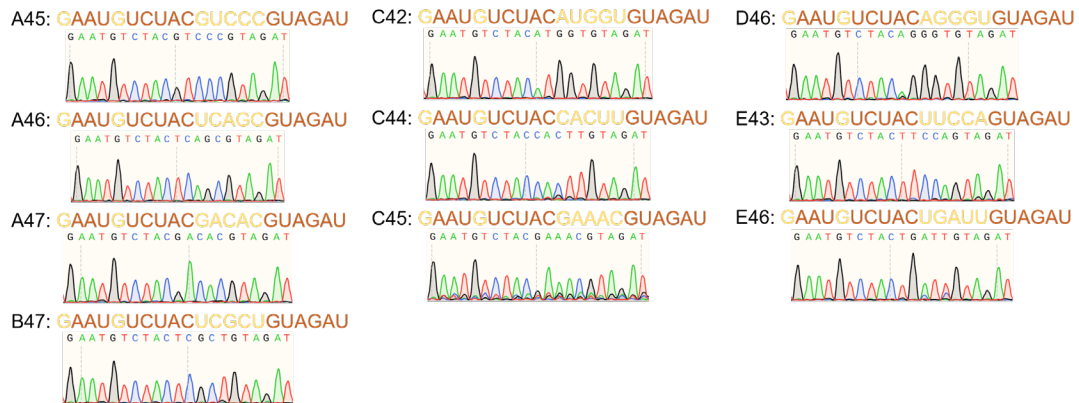


Supplementary Fig. 2 | Sequences of single-part-mutated crRNAs obtained through Sanger sequencing. These mutants were identified during high-throughput screening for their strong ability to repress *gfp* expression.

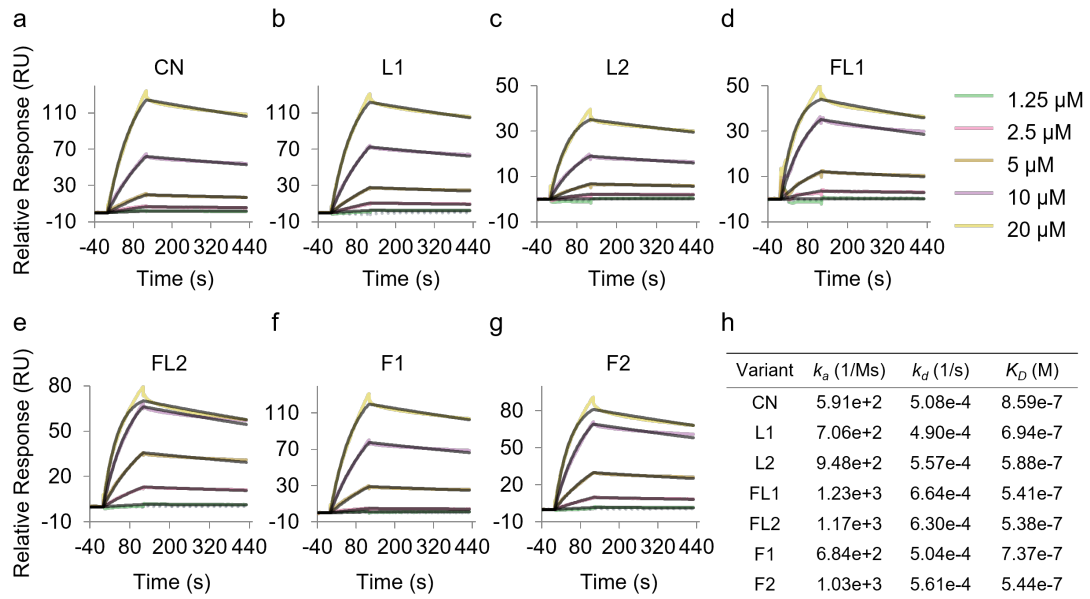
Multi-part-mutated crRNAs (CAAUGUCUACNNNNNGUAGAU):



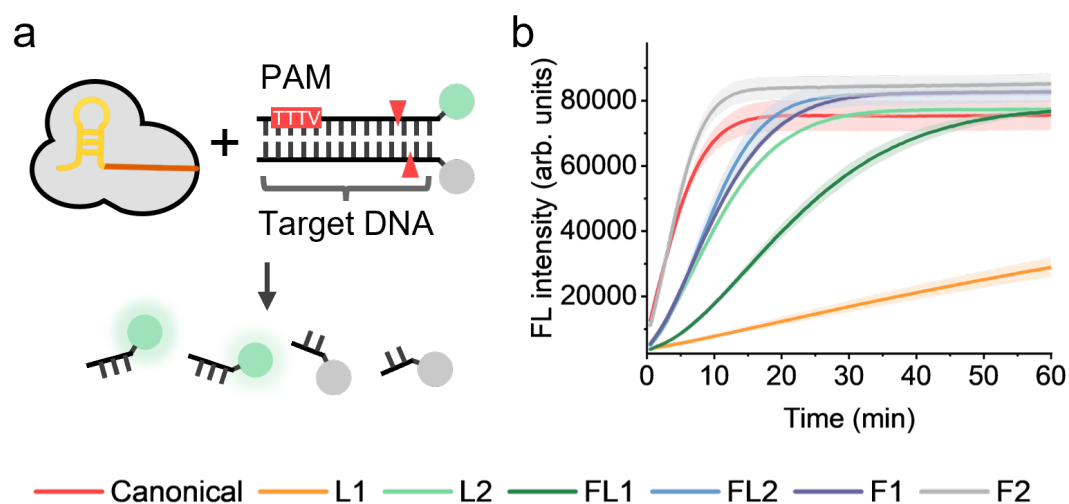
Multi-part-mutated crRNAs (GAAUGUCUACNNNNNGUAGAU):



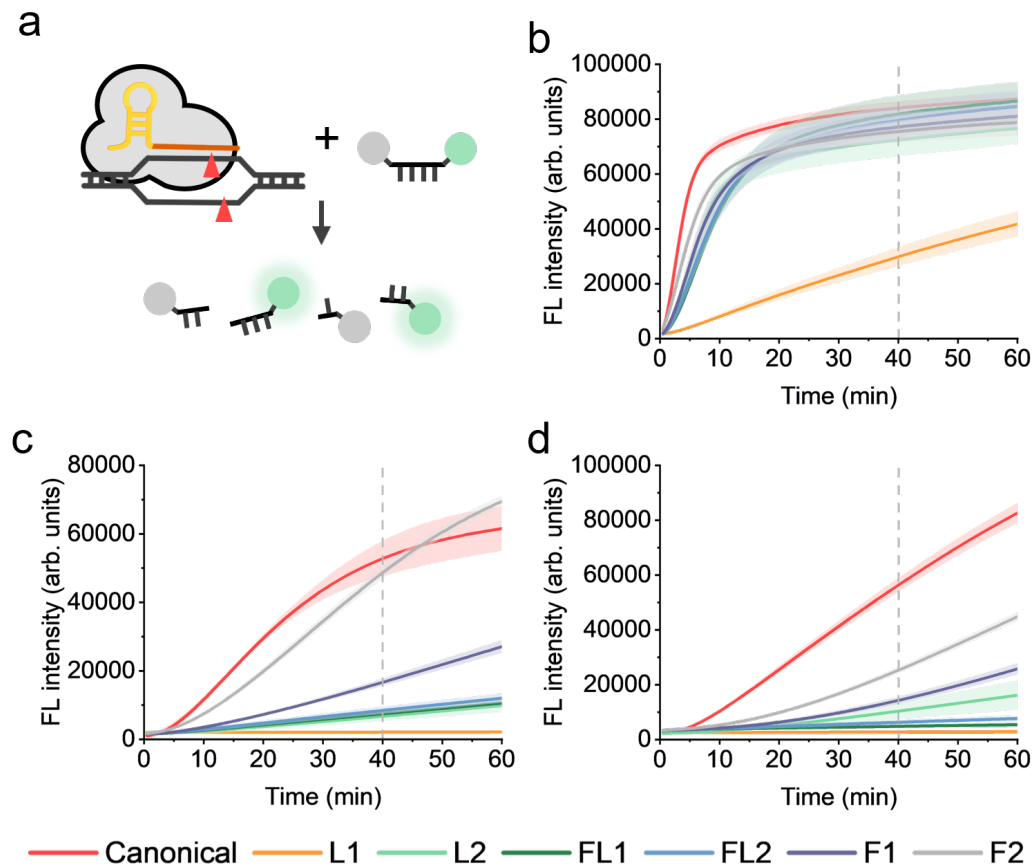
Supplementary Fig. 3 | Sequences of multi-part-mutated crRNAs obtained through Sanger sequencing. These mutants were identified during high-throughput screening for their strong ability to repress *gfp* expression.



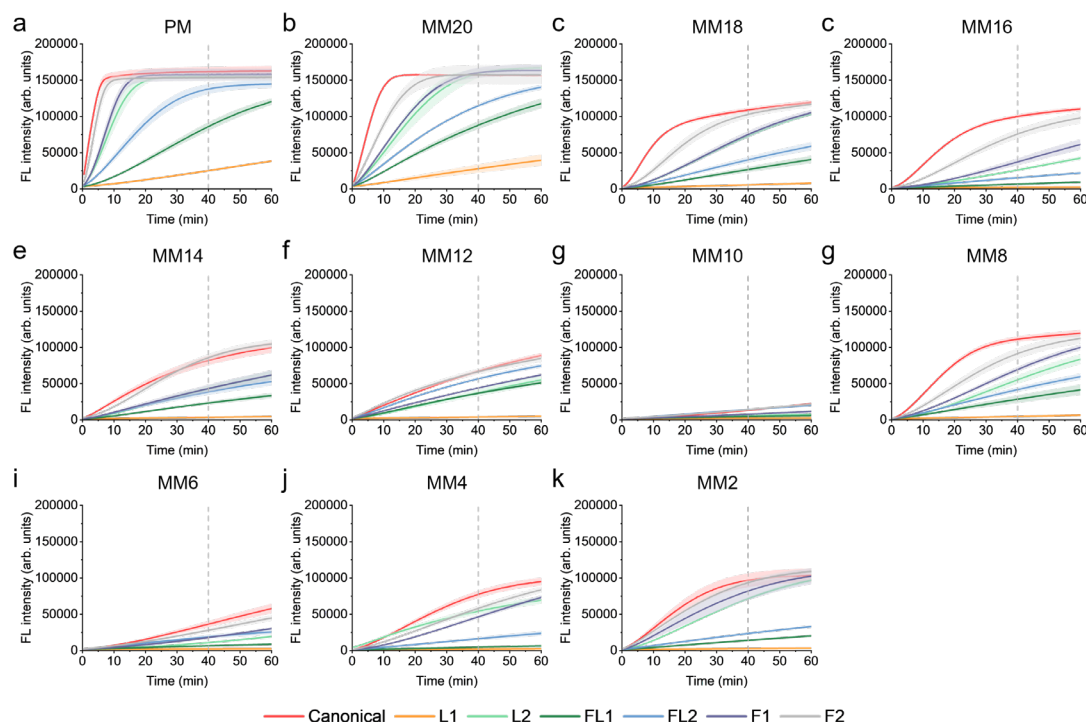
Supplementary Fig. 4 | SPR-based analysis of crRNA–Cas12a binding affinity. a–g, Binding affinity between Cas12a and crRNAs. Surface plasmon resonance (SPR) was used to examine the binding interactions between Cas12a and the canonical crRNA (a), as well as six crRNA mutants: L1 (b), L2 (c), FL1 (d), FL2 (e), F1 (f), and F2 (g). **h,** Kinetic parameters of crRNA–Cas12a binding, including the association rate constant (k_a), dissociation rate constant (k_d), and equilibrium dissociation constant (K_D), derived from SPR analysis. Source data are provided as a Source Data file.



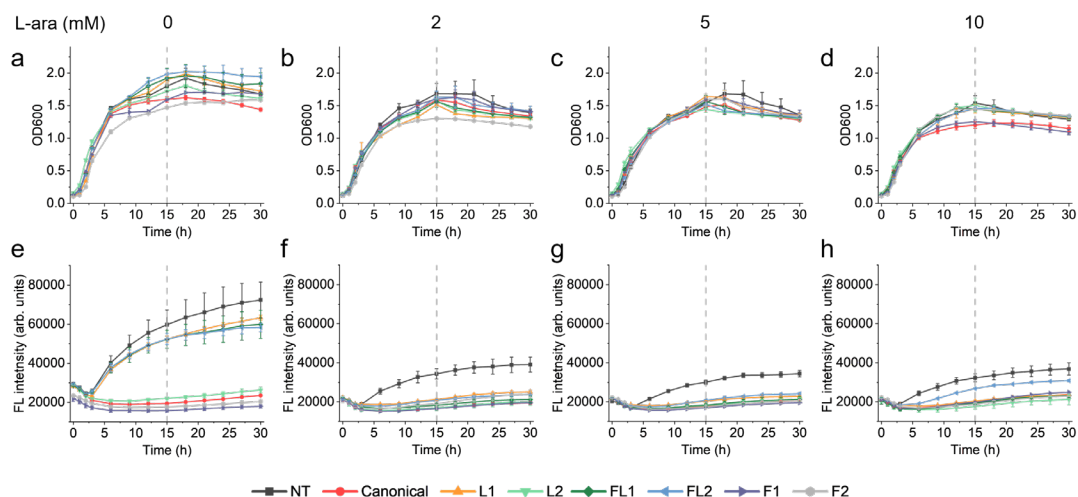
Supplementary Fig. 5 | Fluorescent characterization of *cis*-cleavage. **a**, Schematic of *cis*-cleavage fluorescence characterization. The FAM-labeled non-target strand and BHQ-labeled target strand were pre-annealed to form a double-stranded substrate, which emits fluorescence upon cleavage. **b**, Real-time fluorescence signals of Cas12a *cis*-cleavage activity guided by canonical and direct repeat (DR)-mutated crRNAs in rCutSmart buffer. Shaded areas around the curves represent the standard deviation (SD) of three independent experiments. Source data are provided as a Source Data file.



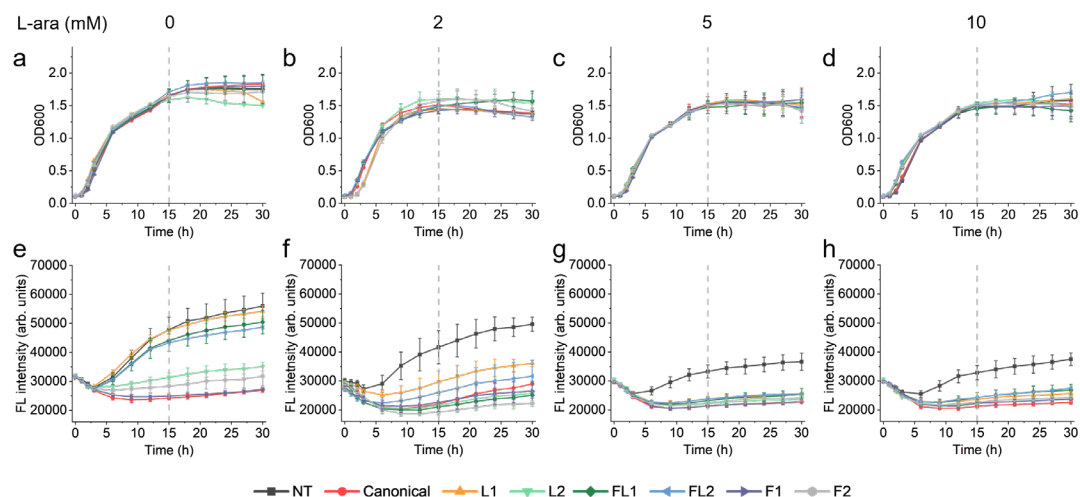
Supplementary Fig. 6 | Fluorescent characterization of *trans*-cleavage. **a**, Schematic of *trans*-cleavage fluorescence characterization. A FAM- and BHQ-labeled probe was employed as *trans*-cleavage substrate, emitting fluorescence upon cleavage. **b-d**, Real-time fluorescence signals of Cas12a *trans*-cleavage activity guided by canonical and DR-mutated crRNAs, with target dsDNA activators at concentrations of 10 nM (**b**), 1 nM (**c**) and 100 pM (**d**). Shaded areas around the curves represent the standard deviation (SD) of three independent experiments. Source data are provided as a Source Data file.



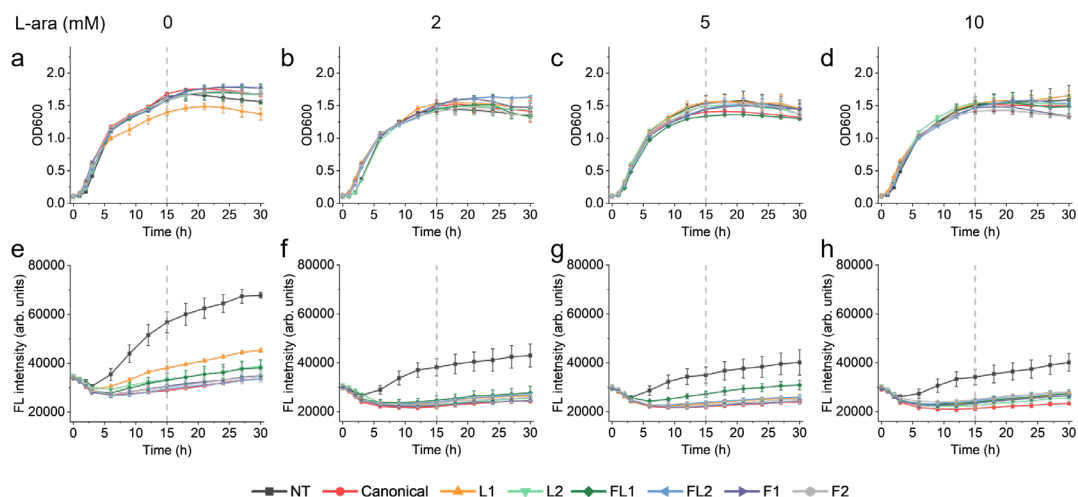
Supplementary Fig. 7 | Real-time fluorescence signals of Cas12a *trans*-cleavage activated by target dsDNA containing double-point mutations. **a**, *Trans*-cleavage signal of Cas12a activated by perfectly matched target dsDNA. **b-k**, *Trans*-cleavage signal of Cas12a activated by dsDNA containing double-point mutations. Each consecutive 2 bp within the protospacer was mutated, from MM20 to MM2, with the mutation sites progressively approaching the PAM (Fig. 2e). Shaded areas around the curves represent the standard deviation (SD) of three independent experiments. Source data are provided as a Source Data file.



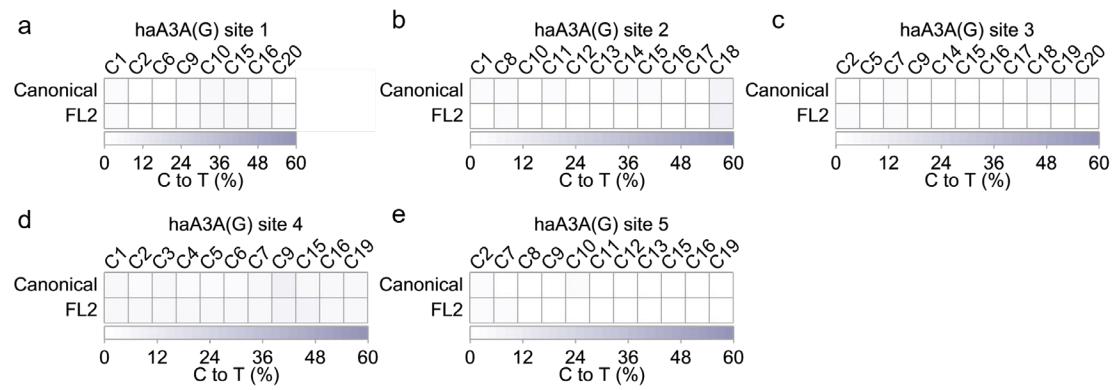
Supplementary Fig. 8 | Growth curves and fluorescence levels with dCas12a targeting *gfp* spacer 1 (corresponding to Fig. 3b, RBS0). **a-d**, Growth curves with dCas12a under the control of RBS0 targeting *gfp* spacer 1, induced with 0 (**a**), 2 (**b**), 5 (**c**), and 10 (**d**) mM l-ara. **e-h**, Fluorescence intensity with dCas12a controlled by RBS0 targeting *gfp* spacer 1, induced with 0 (**e**), 2 (**f**), 5 (**g**), and 10 (**h**) mM l-ara. Canonical and DR-mutated crRNAs were used to guide dCas12a. Data are shown as mean $\pm$ s.d. for $n = 3$ biologically independent samples. Source data are provided as a Source Data file.



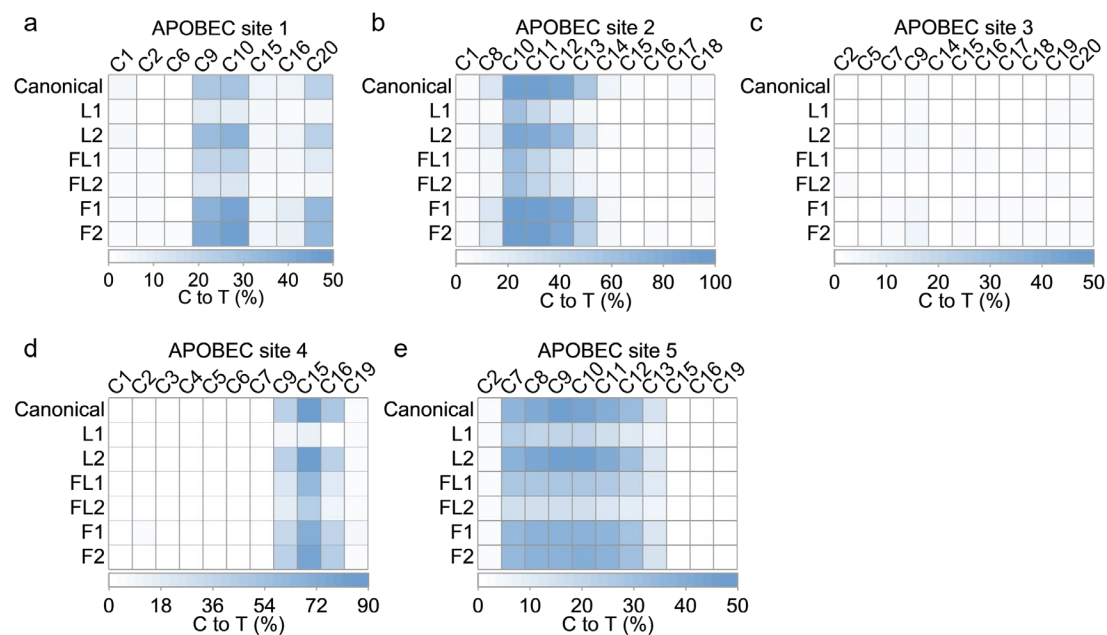
Supplementary Fig. 9 | Growth curves and fluorescence levels with dCas12a targeting *gfp* spacer 1 (corresponding to Fig. 3b, RBS33; Fig. 3c; and Fig. 3d, X-axis). a-d, Growth curves with dCas12a under the control of RBS33 targeting *gfp* spacer 1, induced with 0 (a), 2 (b), 5 (c), and 10 (d) mM l-ara. e-h, Fluorescence intensity with dCas12a controlled by RBS33 targeting *gfp* spacer 1, induced with 0 (e), 2 (f), 5 (g), and 10 (h) mM l-ara. Canonical and DR-mutated crRNAs were used to guide dCas12a. Data are shown as mean $\pm$ s.d. for $n=3$ biologically independent samples. Source data are provided as a Source Data file.



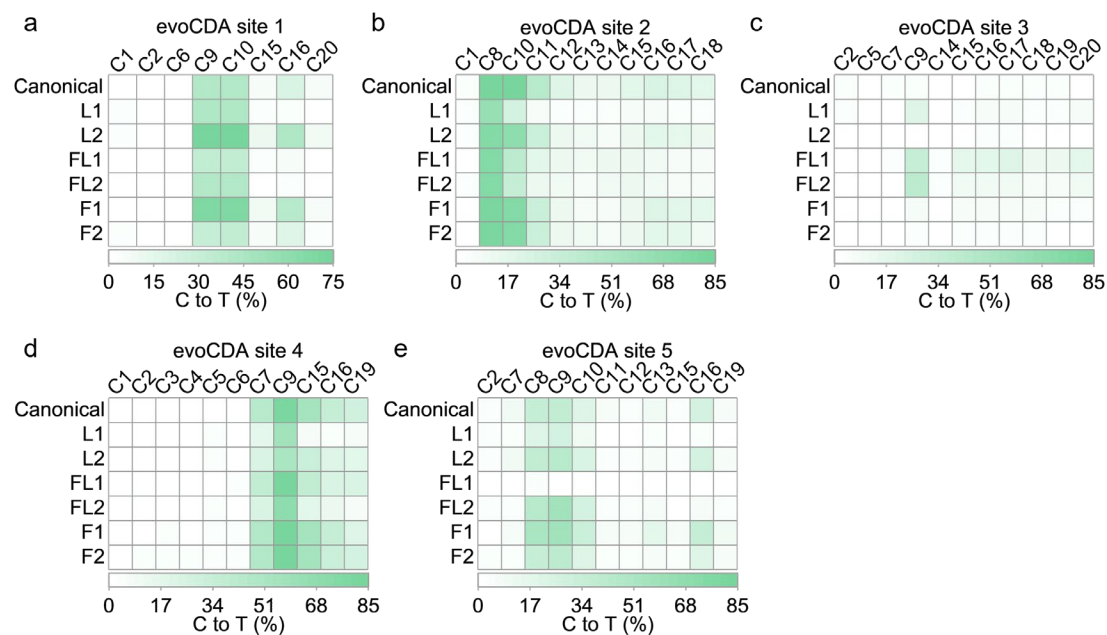
Supplementary Fig. 10 | Growth curves and fluorescence levels with dCas12a targeting *gfp* spacer 2 (corresponding to Fig. 3d, Y-axis). a-d, Growth curves with dCas12a under the control of RBS33 targeting *gfp* spacer 2, induced with 0 (a), 2 (b), 5 (c), and 10 (d) mM l-ara. e-h, Fluorescence intensity with dCas12a controlled by RBS33 targeting *gfp* spacer 2, induced with 0 (e), 2 (f), 5 (g), and 10 (h) mM l-ara. Canonical and DR-mutated crRNAs were used to guide dCas12a. Data are shown as mean $\pm$ s.d. for $n = 3$ biologically independent samples. Source data are provided as a Source Data file.



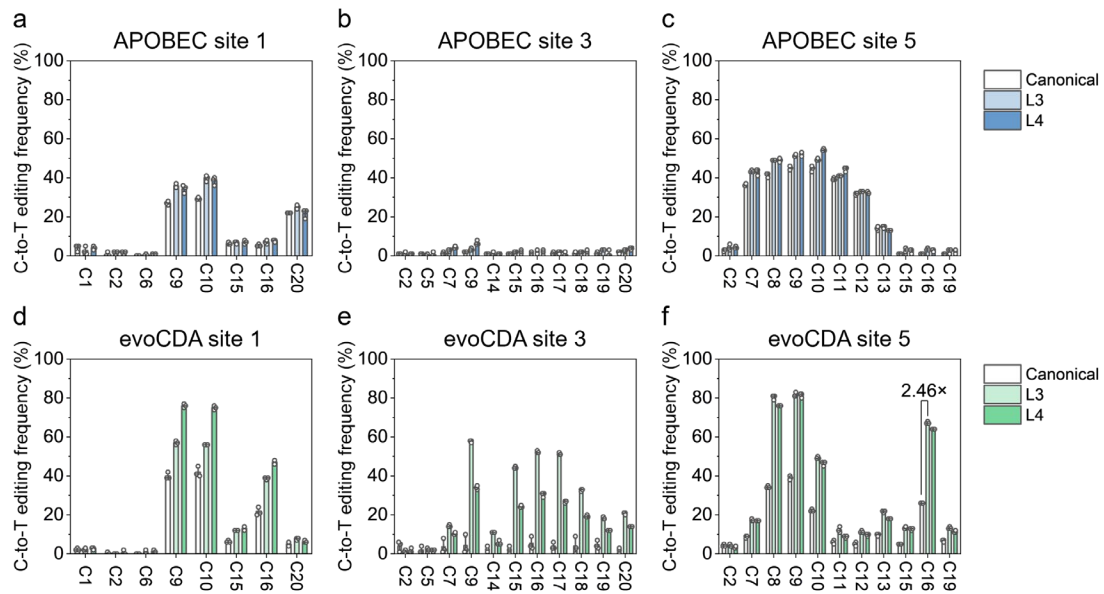
Supplementary Fig. 11 | C to T editing efficiency derived from haA3Aa-dCas12a-UGI. Canonical and crRNA mutant FL2 were used to guide the base editor. Multiple sites (**a-e**) were assessed to validate the editing efficiency. Data represent results from three independent biological replicates (n = 3). Source data are provided as a Source Data file.



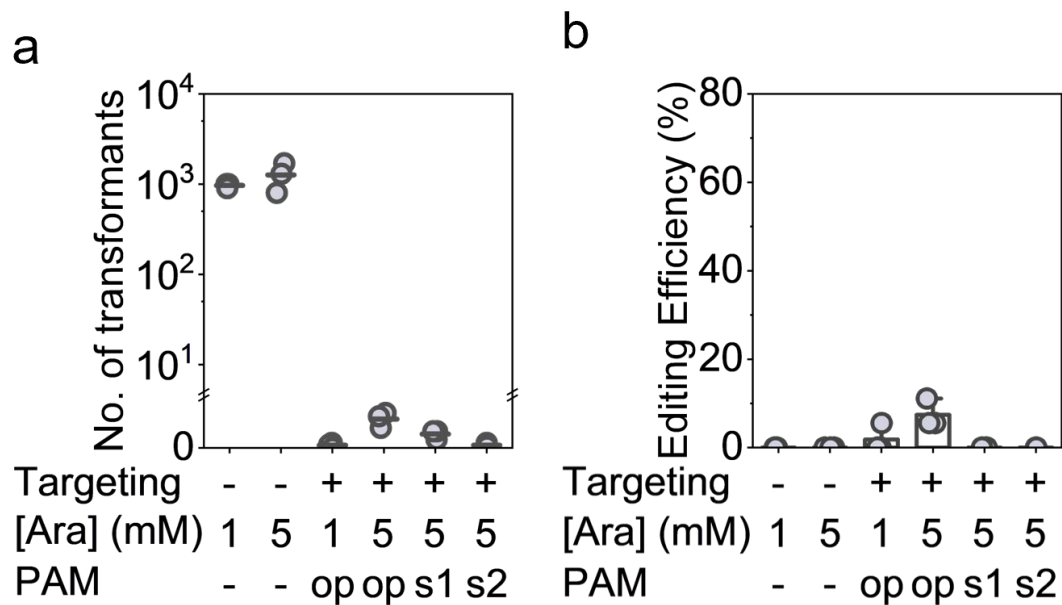
Supplementary Fig. 12 | C to T editing efficiency derived from APOBEC-dCas12a-UGI. Canonical and DR-mutated crRNAs were used to guide the base editor. Multiple sites (**a-e**) were assessed to validate the editing efficiency. Data represent results from three independent biological replicates ($n = 3$). Source data are provided as a Source Data file.



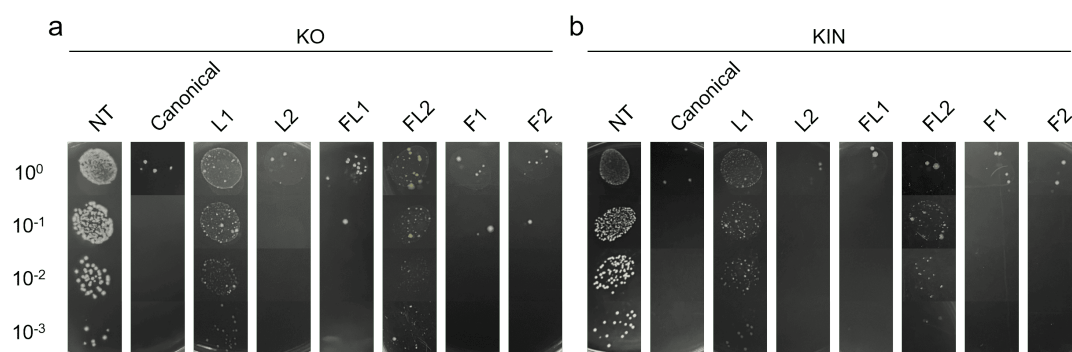
Supplementary Fig. 13 | C to T editing efficiency derived from evoCDA-dCas12a-UGI. Canonical and DR-mutated crRNAs were used to guide the base editor. Multiple sites (**a-e**) were assessed to validate the editing efficiency. Data represent results from three independent biological replicates ($n = 3$). Source data are provided as a Source Data file.



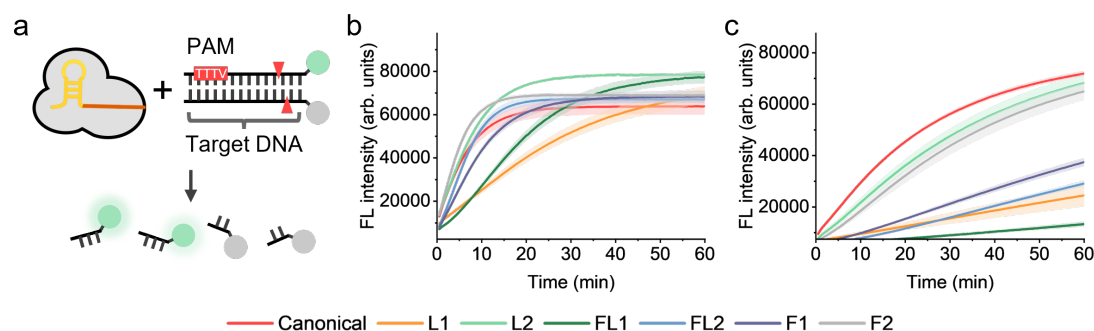
Supplementary Fig. 14 | C to T editing efficiencies guided by L3 and L4. **a-c,** Editing efficiencies of APOBEC-dCas12a-UGI guided by L3 and L4 across multiple sites. **d-f,** Editing efficiencies of evoCDA-dCas12a-UGI guided by L3 and L4 at the same sites. Data are shown as mean $\pm$ s.d. for $n = 3$ biologically independent samples. Source data are provided as a Source Data file.



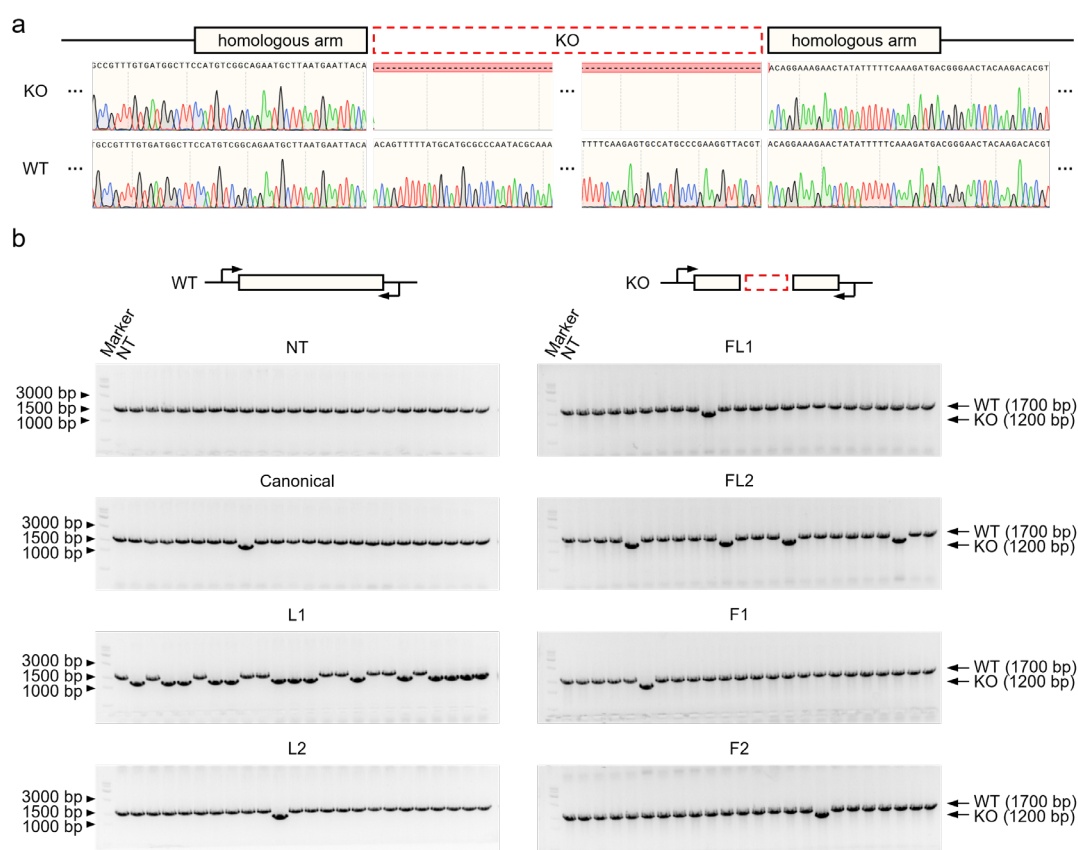
Supplementary Fig. 15 | Reduced Cas12a expression and targeting of suboptimal PAM sites do not enhance transformation (a) or editing (b) efficiency in CRISPR-mediated homologous recombination. Each dot for the transformations represents a single biological replicate (n=3). NT indicates non-targeting gRNAs. Each dot for the editing efficiencies represents the average of 16 colonies screened from one biological replicate (n=3). Bars indicate the mean of the dots. op, optimal PAM (TTTC). s1, suboptimal PAM (CTTA). s2, suboptimal PAM (TCCA). Source data are provided as a Source Data file.



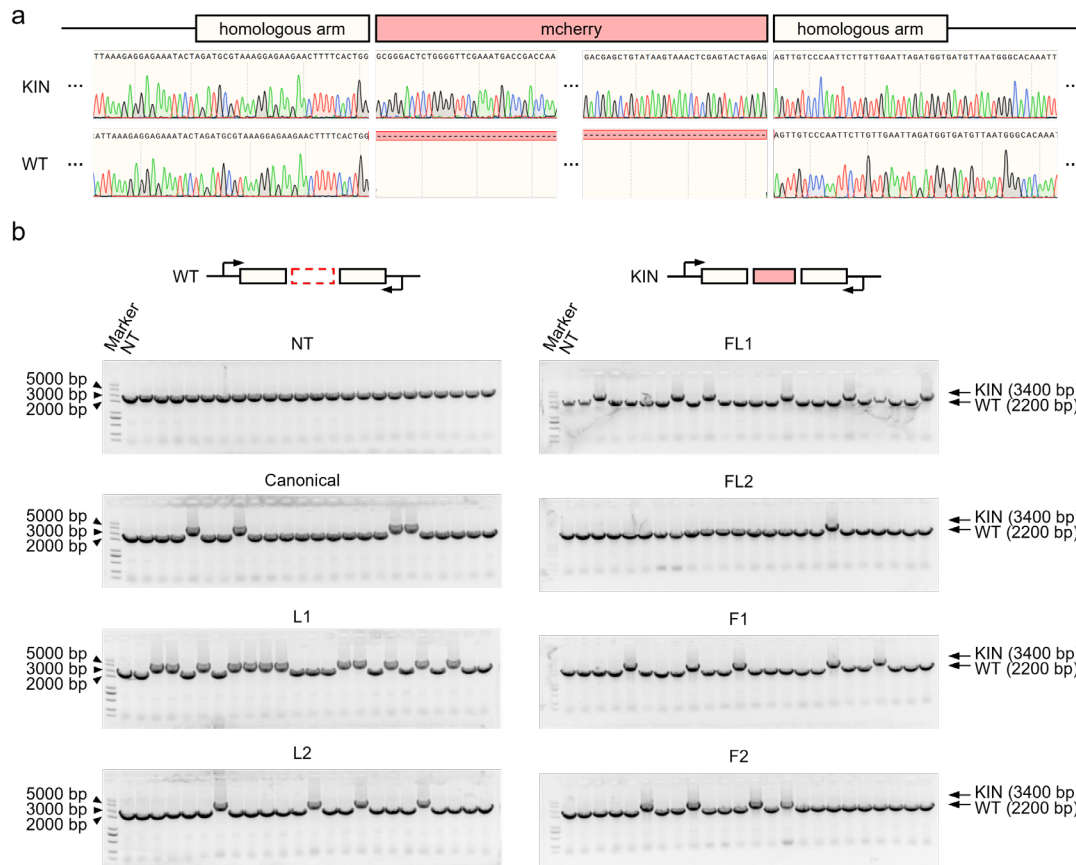
Supplementary Fig. 16 | Transformants of *E. coli* NEB 10-beta-*gfp* strains derived from genome editing assay. a, Transformants from the knockout assay. **b,** Transformants from the knock-in assay. Representative images of transformants are derived from a single experiment without replication. Source data are provided as a Source Data file.



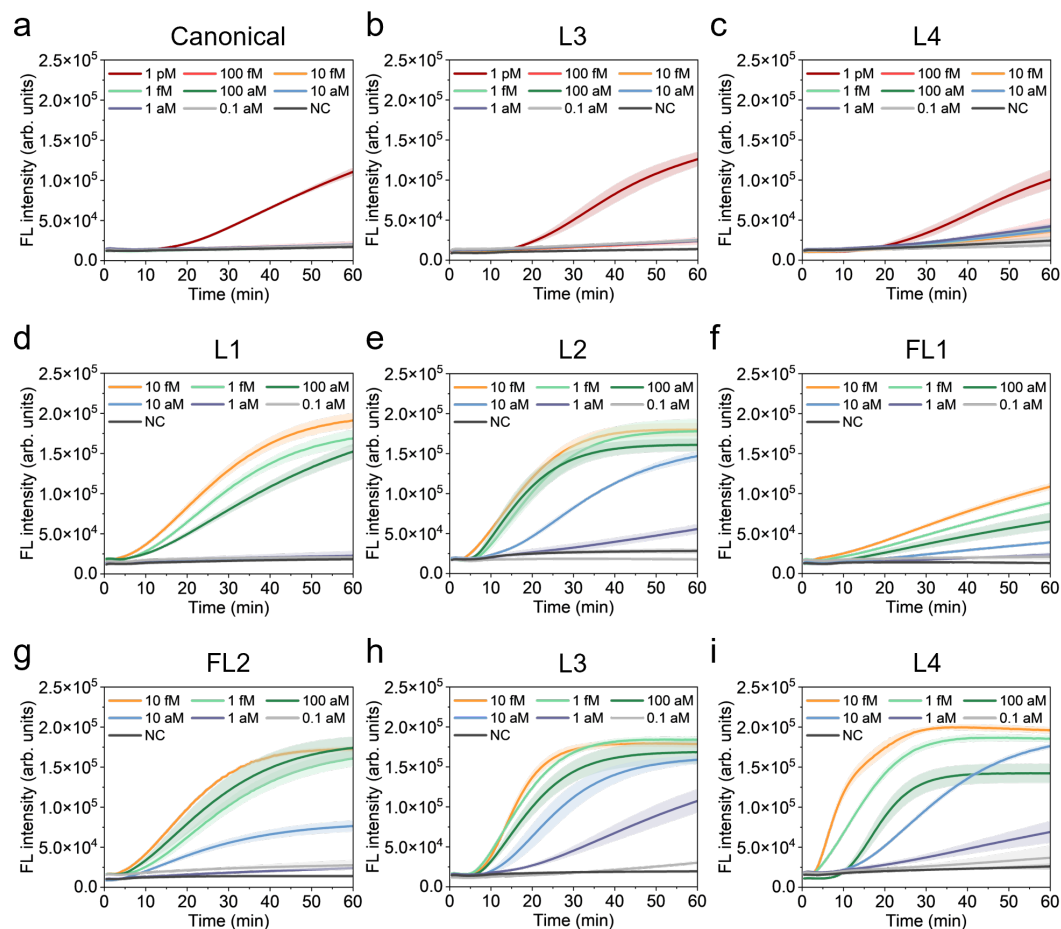
Supplementary Fig. 17 | Fluorescent characterization of in vitro *cis*-cleavage targeting the site used in CRISPR-mediated homologous recombination (corresponding to Fig. 4). **a**, Schematic of *cis*-cleavage fluorescence characterization. The FAM-labeled non-target strand and BHQ-labeled target strand were pre-annealed to form a double-stranded substrate, which emits fluorescence upon cleavage. **b-c**, Real-time fluorescence signals of Cas12a *cis*-cleavage activity guided by canonical and DR-mutated crRNAs in rCutSmart buffer (**b**) and Buffer-1 (**c**). Shaded areas around the curves represent the standard deviation (SD) of three independent experiments. Source data are provided as a Source Data file.



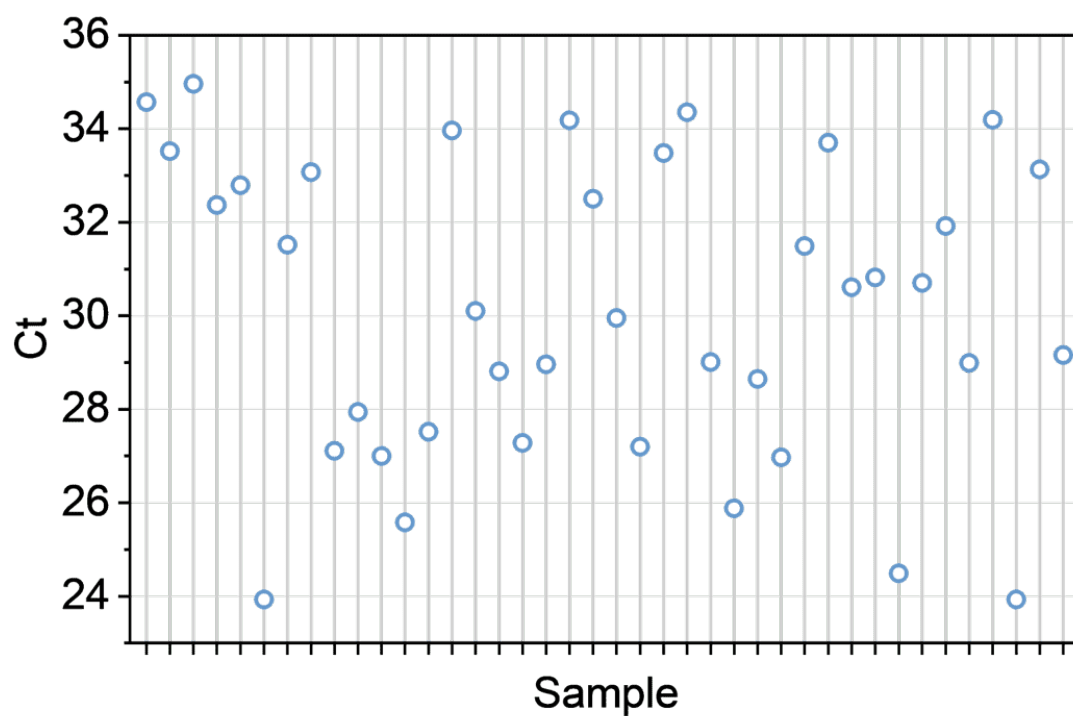
Supplementary Fig. 18 | 500 bp knockout via CRISPR-mediated homologous recombination. **a**, Sanger sequencing confirms the genome sequences of the knockout strain compared to the wild-type strain. **b**, Amplicon length used to assess editing efficiency were obtained using primers annealing outside the homology arms. The presence of a 1700 bp fragment indicates no knockout, while a 1200 bp fragment represents successful knockout. Source data are provided as a Source Data file.



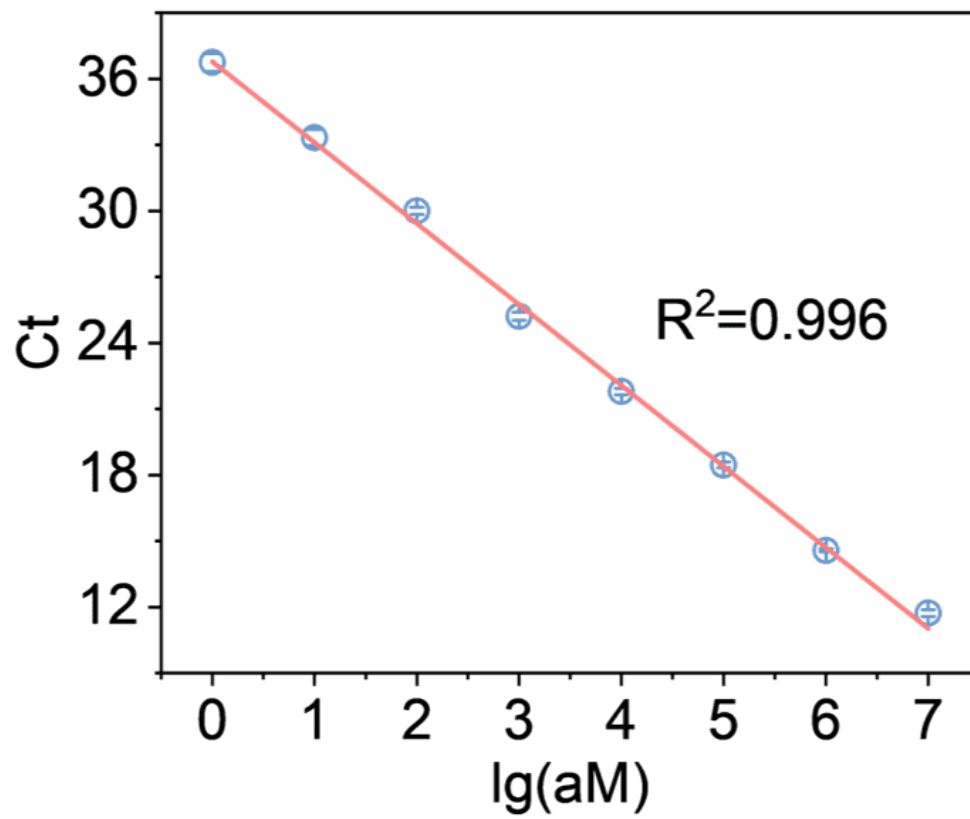
Supplementary Fig. 19 | 1200 bp knock-in via CRISPR-mediated homologous recombination. **a**, Sanger sequencing confirms the genome sequences of the knock-in strain compared to the wild-type strain. **b**, Amplicon length used to assess editing efficiency were obtained using primers annealing outside the homology arms. The presence of a 2600 bp fragment indicates no knock-in, while a 3800 bp fragment represents successful knock-in. Source data are provided as a Source Data file.



Supplementary Fig. 20 | LoD analysis for *Mycoplasma pneumoniae* detection using canonical and DR-mutated crRNAs. Canonical crRNA (a), L3 (b), L4 (c), L1 (d), L2 (e), FL1 (f), FL2 (g), F1 (h), and F2 (i) were employed to identify the synthetic P1 adhesin gene in the one-pot RPA-CRISPR-Dx. Shaded areas around the curves represent the standard deviation (SD) of three independent experiments. Source data are provided as a Source Data file.



Supplementary Fig. 21 | Ct values of *M. pneumoniae*-positive clinical samples. The data shown in the figure are consistent with Supplementary Table 4. Source data are also provided as a Source Data file.



Supplementary Fig. 22 | Standard curve of Ct values versus concentrations of the synthetic P1 adhesin gene in qPCR detection. The curve was further used for quantification of *M. pneumoniae* in clinical samples. Data are shown as mean $\pm$ s.d. for $n = 3$ biologically independent samples. Source data are provided as a Source Data file.

References

1. Hardegger, D., Nadal, D., Bossart, W., Altwegg, M. & Dutly, F. Rapid detection of *Mycoplasma pneumoniae* in clinical samples by real-time PCR. *J. Microbiol. Methods* **41**, 45–51 (2000).
2. Shi, K. *et al.* A CRISPR-Cas autocatalysis-driven feedback amplification network for supersensitive DNA diagnostics. *Sci. Adv.* **7**, eabc7802.